Predicting Veterinarians Licensure Exam Success

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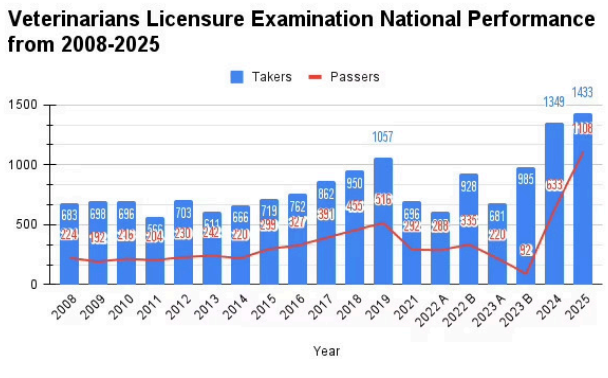

Only 3 out of 10 exam takers will pass the licensure examination for Veterinary Medicine

The Business Problem

VLE volatility creates a real market risk: unpredictable pass rates weaken institutional reputation, increase pressure on student outcomes, and can affect long-term enrollment and stakeholder confidence. Early intervention turns that risk into an opportunity to improve pass rates, protect the brand, and deliver stronger financial and reputational returns.

Low Licensure Passing Rates

Graduates failing Veterinarians licensure exam contributes to underemployment rate in the Philippines

Late Identification

At-risk students identified too late for effective remediation strategies to take effect

No Data System

Current student support lacks data-driven intelligence to guide resource allocation effectively

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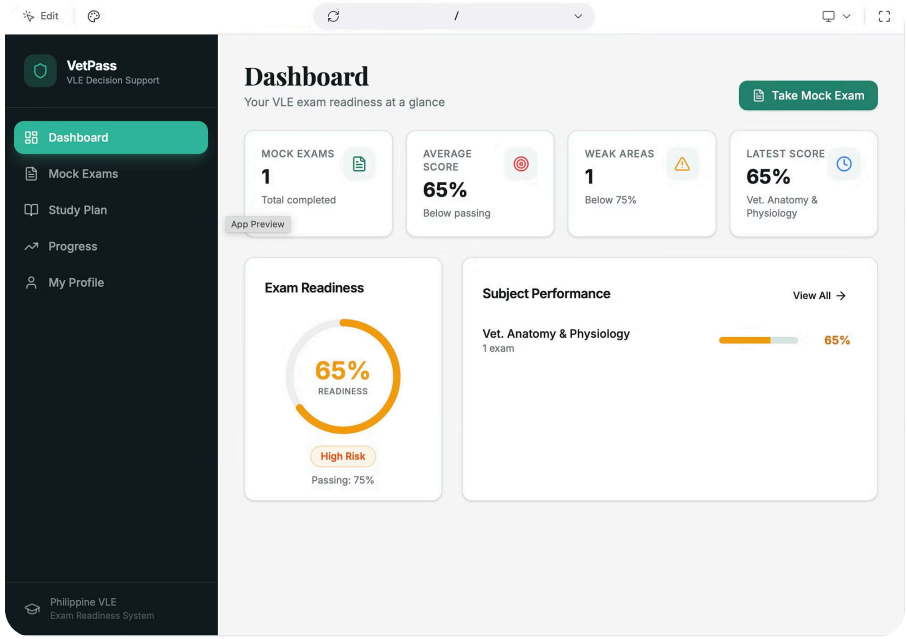
Our Approach

AI-Powered Early Warning System

Machine learning leverages mock exam data to predict licensure outcomes with strong accuracy. The model assigns dynamic risk scores to each student, enabling early identification and targeted interventions through simple LMS integration.

Seamless integration into existing learning management systems



Mock Exam Data

Historical performance as predictive input

ML Model

Risk score calculation in real-time

Early Warning

Proactive alerts for at-risk students

Targeted Support

Personalized interventions drive outcomes

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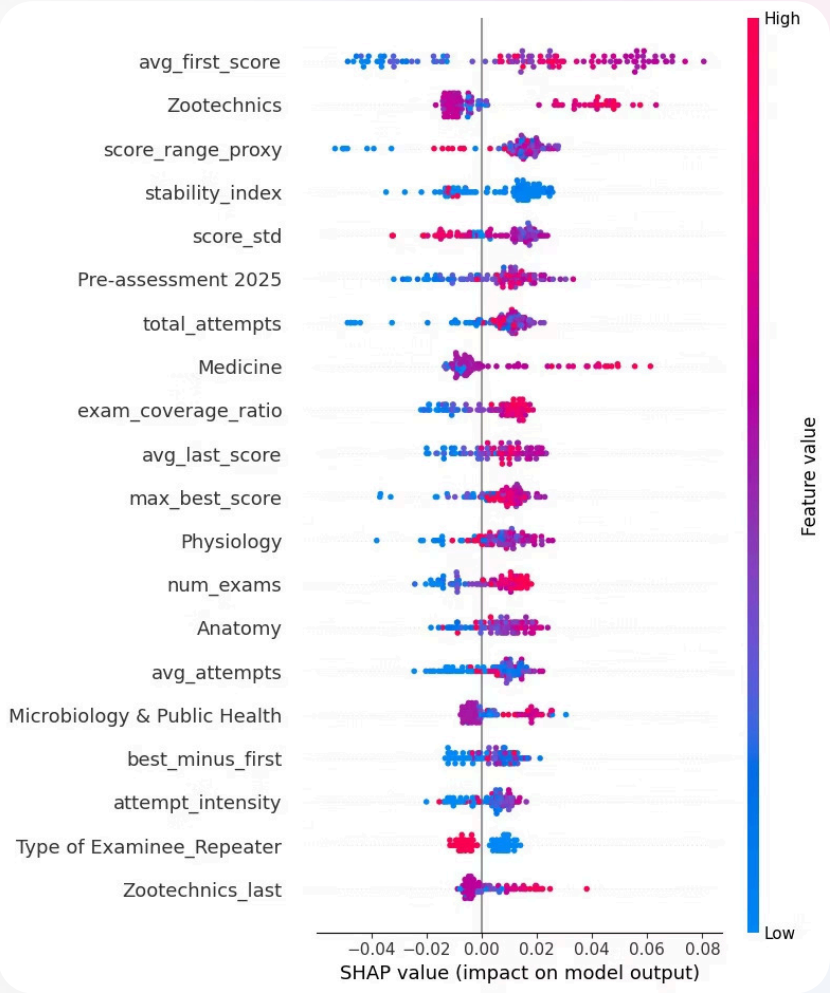

Key Findings

Analysis of mock exam data revealed critical insights into student success predictors and performance patterns.

1

Mock Exam Performance

Historical mock exam scores (avg_first_score) emerged as the strongest predictor of eventual licensure success, with consistent correlation across multiple cohorts

2

Consistency Matters

Performance trends and improvement (stability_index) trajectories provide additional predictive value beyond single test scores

3

Subject-Specific Impact

Certain subjects—particularly Zootechnics, Medicine, Physiology, Anatomy and Microbiology & Public Health —show distinct influence on overall outcome probabilities

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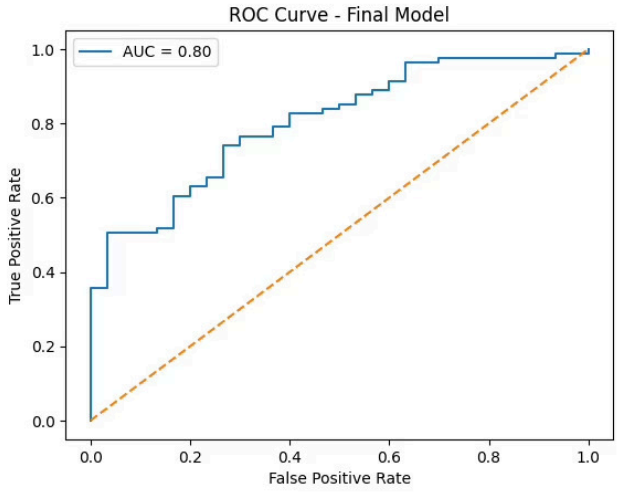
Results & Business Value

Strong Predictive Performance

The machine learning model (Random Forest) achieves ROC-AUC of approximately 0.80, accurately identifying the majority of students who will pass licensure exams. This enables institutions to focus resources where they matter most.

- Majority of passing students correctly identified early
- Early intervention window creates opportunity for improvement
- Model continuously improves with additional data



0.80

ROC-AUC Score

Strong predictive performance across validation cohorts

85%

Passing Student Identification

Accurately flags majority of successful candidates

2-3x

Intervention Window

Early warning enables timely remediation strategies

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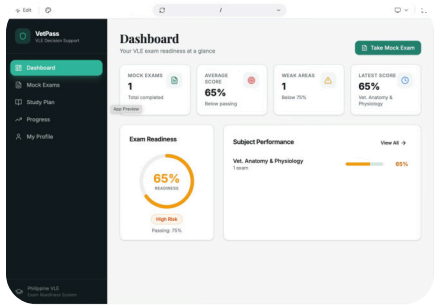

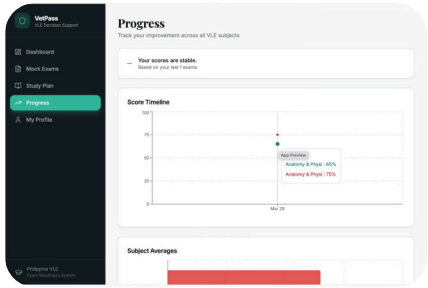

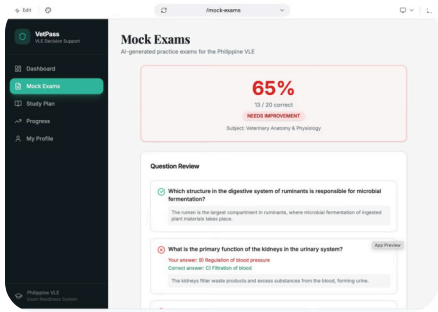

Strategic Recommendations

Deploy Early Warning Dashboard

Implement real-time student risk monitoring accessible to faculty and academic advisors through existing LMS infrastructure

Targeted Review Programs

Design structured remediation pathways for at-risk learners with subject-specific focus and adaptive pacing

AI-Driven Study Plans

Generate personalized learning recommendations based on individual performance gaps and improvement trajectories

Continuous Model Retraining

Refresh predictive models with each new cohort to maintain accuracy and adapt to changing patterns

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Risks & Considerations

While the predictive model delivers strong results, successful implementation requires careful attention to potential challenges and ethical considerations.

Data Bias & Representativeness

Model performance depends on training data quality. Biases in historical data may propagate into predictions if not addressed through careful validation and fairness testing

Over-Reliance on Predictions

Automated risk scores should inform—not replace—human judgment. Faculty expertise remains essential for nuanced student support decisions

Human Validation Required

Faculty oversight ensures predictions align with observed student behavior. Regular calibration meetings maintain model relevance and trust

Data Privacy & Compliance

Student data handling must comply with Data Privacy Act of the Philippines. Clear consent protocols and data security measures are mandatory

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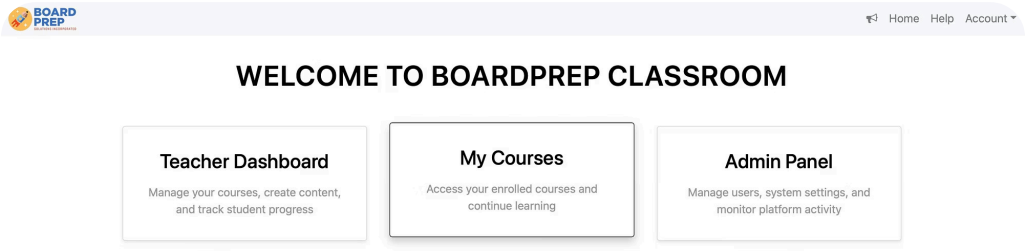
Next Steps & ROI Projection

Implementation Roadmap

Start with a controlled pilot in BoardPrep Classroom to validate model performance and gather user feedback. Expand to partner universities through B2B licensing agreements once results are proven.

Expected Outcomes

- 5–10% increase in passing rates within first year
- Higher enrollment through improved institutional ranking
- Premium AI features as competitive differentiator



Q1: Pilot Launch

Learning Management System deployment with >500 students

Q2: Partner Onboarding

Expand to 3-5 university partners for validation

Q3: Scale Operations

Full rollout across partner network with premium features

Q4: Market Leadership

Establish as industry standard for student success

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